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Research Paper

Preparation and characterization of porous and stable sodium- and potassium-based alkali activated material (AAM)

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ABSTRACT

The aim of this work is to produce highly porous and stable alkali-activated material (AAM) prepared from two combinations of sodium (Na)- and potassium (K)-based alkali solutions ($NaOH/Na_2SiO_3$ and KOH/K_2SiO_3). The reactive metakaolin as precursor and AAM were characterized using X-ray diffraction spectroscopy (XRD), X-ray fluorescence spectroscopy (XRF), aluminum nuclear magnetic resonance spectroscopy (^{27}Al MS-NMR), diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS), field emission scanning electron microscopy (FESEM), compressive strength measurement and Brunauer–Emmett–Teller (BET) surface analysis. The porosity of the AAMs were increased by using hydrogen peroxide and sodium percarbonate as foaming agents. Characterization results showed the viscosity of the K-AAM paste was 70% lower than that of the Na-AAM paste. However, the volume of the Na-AAM paste in an air-tight plastic tube was three times higher than that of K-AAM, but the specific surface area (SSA) of K-AAM were 30% higher than those of Na-AAM. In terms of compressive strength, the blank AAM (foaming agent-free) demonstrated the highest strength values: 6.1 MPa for K-AAM and 9.0 MPa for Na-AAM. When the concentration of the foaming agent was increased, the compressive strength of both the materials decreased but were still around 1.0 MPa. The FESEM images of the Na-AAM and K-AAM produced with H_2O_2 indicated the high porosity of materials which were also observed in SSA values of AAM. Furthermore, the XRD data showed that the Na-AAM contained water in hydrate form (halloysite) compared with the K-AAM, suggesting the different polymerization reaction route and speed between these AAM.

1. Introduction

Alkali-activated cements or geopolymers are inorganic polymers which are formed in the reaction between amorphous alumino-silicate raw materials and highly concentrated aqueous solution of alkali compounds (e.g., silicates and hydroxides) producing stable three dimensional polymeric structures (Font et al., 2020). The sodium (Na)- and potassium (K)-based alkaline solutions have been used as alkaline activators in the manufacture of concrete; however, only a few studies have compared the effect of these elements when used in other applications, such as in water treatment, in which the porosity plays a major role. Bai and Colombo (2018) classified the pore formation processes into (a) direct foaming, (b) sacrificial templating, (c) emulsion templating, (d) additive manufacturing, and (e) replica templating and their combinations (Bai and Colombo, 2018). AAM foams display high porosity, excellent stability at high temperatures, low thermal conductivity and good mechanical properties (Lecomte et al., 2003; Hlaváček et al., 2015; Palmero et al., 2015; Hemra and Aungkavattana, 2016; Novais et al., 2016a). (Al-Majidi et al., 2016)AAM foams have been less commonly used as adsorbents and filters (López et al., 2014; Luukkonen et al., 2016; Minelli et al., 2016; Novais et al., 2016b), as catalysts (Zhang et al., 2012; Sharma et al., 2015; Heponiemi et al., 2021) and both as membranes and as a supporting material for membranes (Ge et al., 2015; Bai and Colombo, 2017), in where the porous end-product is highly desired. AAM foams used as building materials focus on different properties (strength, hardness and water repellence) than products aimed to water purification processes (Zhang et al., 2015; Papa et al., 2016).

In the method described by Sanjayan et al. (2015), porous products are formed by dissolving foaming agents in AAM pastes under basic conditions (Sanjayan et al., 2015). AAM pastes may also contain other additives, such as aluminum (Al) or silicon (Si) powders or Si-containing

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compounds (Medri et al., 2013; Prud'homme et al., 2015). As described by Verdolotti et al. (2015), various additives (e.g., foaming agents and surfactants) and various rheology and solidification conditions promote the formation of a porous structure and increase the active surface area of a material(Verdolotti et al., 2015). According to studies by Bae et al. (2009) and Zhou et al. (2017), the pores in a material can be formed by porogens (such as hydrogen peroxide (H_2O_2) or ammonium hydrogen carbonate (NH_4HCO_3)) or templates (such as carbohydrate sugars, polystyrene, or other polymers) (Bae et al., 2009; Zhou et al., 2017).

The focus of this study is to produce porous, stable, AAM and characterize them thoroughly using X-ray diffraction spectroscopy (XRD), X-ray fluorescence spectroscopy (XRF), aluminum nuclear magnetic resonance spectroscopy (^{27}Al MS-NMR), diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS), and field emission scanning electron microscopy (FESEM), compressive strength measurement and Brunauer–Emmett–Teller (BET) surface analysis.

2. Experimental

The metakaolin, which have numerous studies and data available, was used as the precursor and it was prepared from powdered kaolin that was calcined at a high temperature (750 °C). The molar ratios for X_2O/SiO_2, SiO_2/Al_2O_3, X_2O/Al_2O_3, and H_2O/X_2O (X = Na, K) were calculated in order to compare the properties of the Na-based AAM (Na-AAM) and K-based AAM (K-AAM). Rheological tests were performed with fresh paste, after which they were cured at 60 °C for 24 h and 105 °C for 72 h, and characterized by various analyzing techniques. The experimental study comprised:

(a) characterization of kaolin and calcined kaolin i.e., metakaolin by XRF, XRD, DRIFT and ^{27}Al MS-NMR.
(b) calculating the necessary ratios to achieve desired end-product and manufacture the AAM with and without porogens (hydrogen peroxide (H_2O_2) and sodium percarbonate (SPC))
(c) measure the viscosity of freshly prepared AAM pastes and both compressive strength and BET of cured AAM samples.
(d) characterization of AAM by XRD, XRF, DRIFT and ^{27}Al MS-NMR.

2.1. Materials

Alkali solutions were prepared from sodium hydroxide (NaOH; Anal. Normapur, VWR Chemicals, Czech Republic) and potassium hydroxide (KOH) pellets (Emsure®, Merck KGaA, Germany) mixed with sodium silicate solution (Na_2SiO_3, extra pure; Supelco®, Merck KGaA, Germany) and potassium silicate solution (K_2SiO_3; Sateenkaari Perinnetalo. Finland). The Na_2SiO_3 solution consisted of 27% SiO_2, 8.0% Na_2O, and 65% H_2O, whereas the K_2SiO_3 solution consisted of 30.1% SiO_2, 9.4% K_2O, and 60.5% H_2O. Hydrogen peroxide (H_2O_2; 30%*W/V* AnalR NORMAPUR, VWR Chemicals, France) and sodium percarbonate (SPC, experimental pilot product for laboratory tests) were used as foaming agents (porogens) to create micro- and meso-pores in the AAM structure. Washed white kaolin (VWR Chemicals, Belgium) was used to produce metakaolin (i.e., calcined kaolin).

2.2. Methods

The viscosity of the pastes was measured at room temperature by a Brookfield Viscometer DV-II+ Pro EXTRA (model LVDV-II + PX; Brookfield Engineering Laboratories, USA) using the Rheocalc software provided by the manufacturer. The same spindle (63 for NA-AAM and 62 for K-AAM) was used in each test. Each rheological measurement was always done with fresh pasta and the measurements were performed after 1 min when the stirring was stopped. In this way, any partial curing processes were minimized. In numerous studies it has been proved that the water inside the structure can be used as an indicator of the curing process for both concretes and geopolymers (Lura et al., 2006; Hu et al., 2020; Li et al., 2021).

Compressive strength measurements were performed on a Zwick/Roell testControl II-unit v7.62 (type BW91272; Zwick/Roell, Germany) equipped with a Tesproma compression unit (Tesproma, Finland). Compression data were collected using the testXpert II *V*3.6 software. A constant compressive force (10kN) was used in all measurements, and the driving speed was 1 mm/min. The measurement stopped automatically when 30% of the sample was crushed or when the maximum compression (10 MPa) was achieved.

BET measurements were performed using a Micromeritics ASAP 2020 instrument (Micromeritics Instruments, Norcross, GA, USA). The specific surface area (SSA) of the Na-AAM and K-AAM samples were determined from the nitrogen physisorption isotherms measured at −196 °C. The crystalline phase of the pure kaolin, manufactured metakaolin, and AAM samples was determined via the XRD method (PANalytical X'Pert Pro, Almelo, The Netherlands) using monochromatic Cu Kα1 radiation ($\lambda = 1.5406$ Å) at 45 kV and 40 mA with a scan step size of 0.017° and 2θ of 8°–85°. The diffractograms were analyzed using X'Pert Highscore (PANalytical B.V., The Netherlands) and compared with the Powder Diffraction File standards from the International Centre for Diffraction Data (ICDD) (PDF-4+ 2020).

The composition of metal oxides in kaolin and calcined kaolin were determined by an XRF spectrometer (PANanlytical Axios mAX XRF, Almelo, The Netherlands) and were used to calculate the desired AAM ratios. The measurements were performed using loose powders run through transparent Mylar films under He atmosphere.

The microstructure and morphology of the samples were investigated using a Zeiss Sigma FESEM (Carl Zeiss Microscopy GmbH, Jena, Germany). The FESEM images were taken at 5 k*V* at magnifications ranging from 150 to 100 K at the Centre for Material Analysis, University of Oulu.

Solid-state ^{27}Al magic-angle spinning MAS-NMR spectra were recorded on a Bruker Avance III 300 spectrometer operating at 78.24 MHz for ^{27}Al. The samples were packed into 7 mm zirconia rotors and then subjected to a spinning frequency of 7 kHz. The spectra were collected at a pulse tilt angle of 30°, and 2048 scans were accumulated at a repetition rate of 2 s. The chemical shifts were referenced to $Al(NO_3)_3$ at 0 ppm.

DRIFTS was used to investigate the degree of polymerization in the prepared samples. The DRIFT spectra were recorded on a Bruker PMA 50 Vertex 80 V (Bruker, Billerica, MA, USA) equipped with a Harrick Praying Mantis™ diffuse reflection accessory for baseline measurement using KBr. Measurements were conducted at 400–4000 cm^{-1} with a resolution of 4 cm^{-1} and 500 scans/min.

2.3. Preparation of Na-AAM and K-AAM

Kaolin was heated to 750 °C in a ceramic oven (Model L5/11/P320; Nabertherm Gmbh, Germany) at a rate of 5 °C/min. This procedure has been found to provide the optimum conditions for the calcination of kaolin to produce metakaolin (Moodi et al., 2011; Davidovits, 2017). Also, the calcination temperature of 750 °C has been reported to impart the maximum compressive strength (Kong et al., 2008; Ilić et al., 2010; Shi et al., 2011; Tchadjie and Ekolu, 2018). The mass loss during calcination was 12.5%. To confirm that the mass loss was not adsorbed water from air, pure kaolin was dried in oven at 105 °C for 48 h and the mass loss was 1.0%. Therefore, there was an assumption that the mass loss was due to interstitial water in kaolin. AAM was manufactured using two alkali combinations: $NaOH/Na_2SiO_3$ and KOH/K_2SiO_3. The molar ratios in Na-AAM and K-AAM were kept the same. The optimal molar ratios were $X_2O/SiO_2 = 0.3$, $SiO_2/Al_2O_3 = 3.5$, and $X_2O/Al_2O_3 = 1.0$ for the AAM and $H_2O/X_2O = 14.1$ and $X_2O/SiO_2 = 0.7$ for the alkali solution (X = Na, K).

Three different preparations were performed to produce samples (1) with constant alkali and AAM ratios but with different H_2O_2

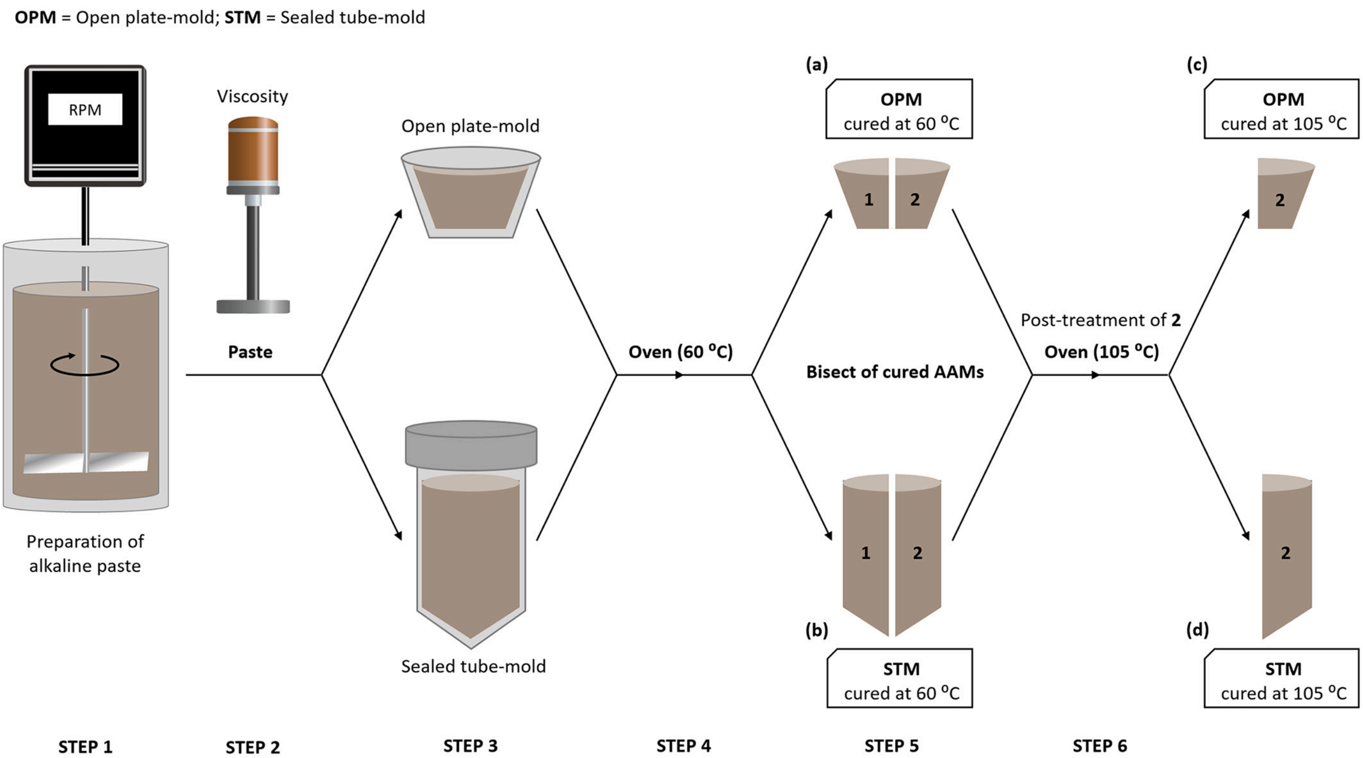

Fig. 1. Preparation of Na-AAM and K-AAM from metakaolin (steps 1–6).

concentrations, (2) with constant alkali ratios and AAM but with different SPC concentrations, and (3) with variable H_2O/X_2O and X_2O/SiO_2 ratios but with constant AAM ratios.

Blank Na-AAM and K-AAM samples were manufactured using steps 1–6 (Fig. 1) with or without a foaming agent. To obtain the desired ratios for the alkali solution, NaOH or KOH pellets were added to the silicate solution and then stirred for 2.5 h.

Porous AAM was prepared using the same process (steps 1–6) shown in Fig. 1. To obtain a certain degree of porosity, different concentrations (0.1–0.5 mass% in 0.1% steps) of either H_2O_2 or powdered SPC (<250 µm) as foaming agent were added. H_2O_2 will be decomposed into water and oxygen gas, forming air bubbles in the paste and thus rendering the AAM structure porous (Eq. (1)) (Petlitckaia and Poulesquen, 2019):

$$2H_2O_2(l) \rightarrow 2H_2O\ (l) + \uparrow O_2(g) \tag{1}$$

Granulated SPC powder was also used as a single blowing agent (porogen) because it dissolves slowly in water, releasing oxygen as it decomposes. The microbubbles released in this case were more homogeneously distributed in the paste than those produced by the H_2O_2 solution (Eqs. (2) and (3)):

$$2Na_2CO_3 \bullet 3H_2O_2 \rightarrow 3H_2O_2 + 4Na^+ + 2CO_3^{2-} \tag{2}$$

$$2H_2O_2 \rightarrow 2H_2O + \uparrow O_2 \tag{3}$$

Pre-made alkali solution was poured into the mixing container (Fig. 1, step 1). Calcined metakaolin powder was added gradually into the alkali solution and mixed at 1000 rpm. The AAM pastes were mixed using IKA RW 20 digital overhead stirrer (IKA®-Werke GmbH & CO.KG, Germany) with an R 1382 three-bladed propeller stirrer (IKA®-Werke GmbH & CO.KG, Germany). In the final stage, mixing was stopped, and the viscosity of the freshly prepared paste was measured (step 2). The paste was divided into two portions; one part was poured into the open mold placed in a sealed plastic bag, whereas the other part was placed in an air-tight plastic tube (step 3). Both subsamples were cured in an oven (Type TS 8056, Termaks, Bergen, Norway) at 60 °C for 24 h (step 4). The cured samples were cut in half (step 5), and the subsamples (plate and tube) were cured in an oven (Modell 600. Memmert GmbH+Co.KG Schwabach. Germany) at 105 °C for 72 h (step 6). This preparation process yields four types of AAM: (a) AAM open plate-mold 60 °C, (b) AAM sealed tube-mold 60 °C, (c) AAM open plate-mold 105 °C, and (d) AAM sealed tube-mold 105 °C.

All samples were weighed to determine the moisture lost during the curing process. The volume of the paste in the air-tight sealed plastic tubes was also measured to determine the paste expansion during the curing process.

Moreover, two different washing methods and washing temperatures influence on the SSA of AAM cured at 105 °C was investigated. The AAMs leach alkaline counter ions Na^+ or K^+ from the material changing the pH of the environment and NaOH or KOH crystal formed on the surface of the material also interfere SSA measurements. Hence, the Na-AAM and K-AAM samples were treated with 1 M acetic acid by refluxing at boiling point and on a laboratory shaker at room temperature. The literature contains some references to the post-processes of alkali-activated materials, including washing with pure water or acid solutions. Hot acid washing processes have also been studied. These methods have been utilized also in this study (Selkälä et al., 2020; Ukrainczyk and Vogt, 2020).

Table 1
XRF analysis results showing the concentrations of the constituents of pure kaolin and calcined kaolin (mass%).

Substance	Na_2O	MgO	Al_2O_3	SiO_2	P_2O_5	SO_3	K_2O	CaO	TiO_2	MnO	Fe_2O_3
Pure kaolin	0.04	0.34	40.42	54.06	1.78	0.08	1.82	0.28	0.07	0.01	0.95
Calcined kaolin	0.36	0.30	41.44	53.13	1.66	0.05	1.69	0.26	0.05	0.01	0.89

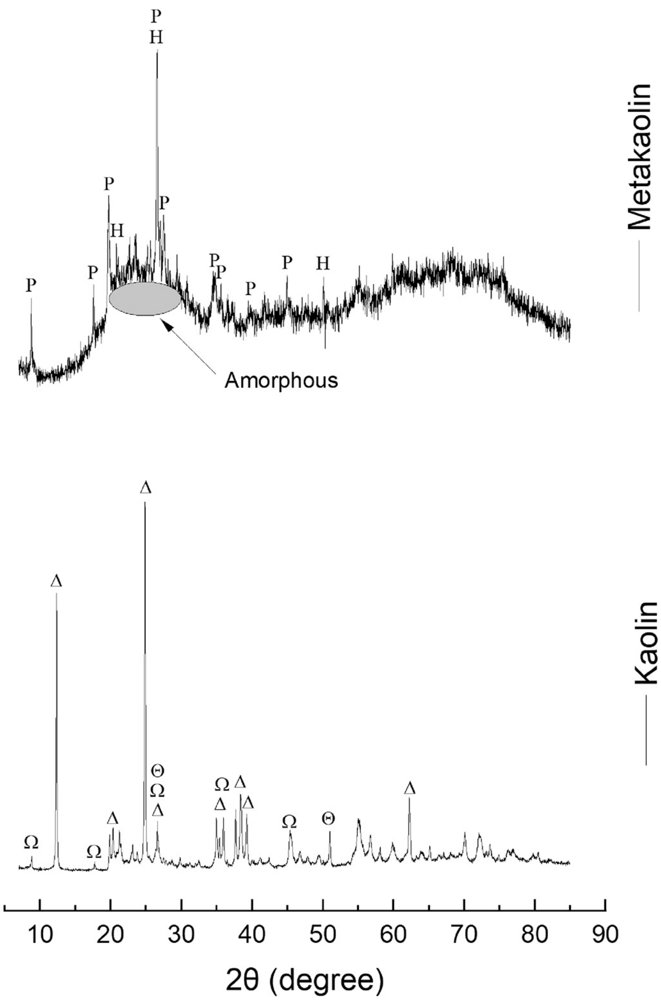

Fig. 2. XRD of kaolin and calcined kaolin (metakaolin). Legend: Δ = aluminum silicate hydroxide (kaolinite), Θ = silicon oxide, Ω = potassium aluminum silicate hydroxide (muscovite), H = hexagonal silicon oxide (quartz), and P = potassium aluminum silicate.

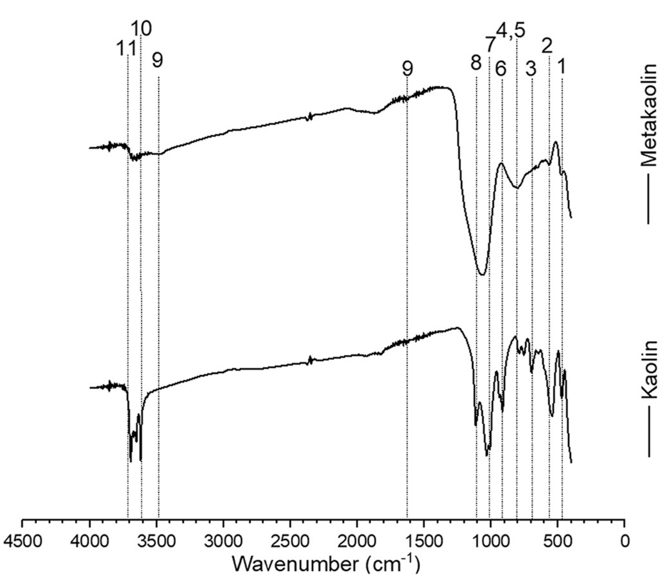

Fig. 3. DRIFT spectra of kaolin and metakaolin.

2.4. Characterization of metakaolin

To obtain the correct AAM ratios, the composition of metal oxides in both kaolin and calcined kaolin were analyzed by XRF. The results are given in Table 1.

XRD measurements were conducted to assess the success of the calcination. The diffractograms are presented in Fig. 2. An amorphous hump is noticeable at 2θ values of 20°–30°, which is typical for amorphous aluminosilicates (Barbosa et al., 2000a).

The XRD reflections of pure kaolin (Fig. 2) were matched with aluminum silicate hydroxide ($Al_2Si_2O_5(OH)_4$; ICDD 04–013-2815), silicon oxide (SiO_2; ICDD 01–070-3755) and potassium aluminum silicate hydroxide ($KAl_2(Si,Al)_4O_{10}(OH)_2$; ICDD 00–058-2037). For the calcined kaolin substances, the reflections were matched with hexagonal quartz (SiO_2; ICDD 01–070-3755) and potassium aluminum silicate ($KAl_3Si_3O_{11}$; ICDD 00–046-0741). The major hexagonal SiO_2 (quartz) reflections in the metakaolin diffractogram are found at 2θ values of 20.9 and 26.6.

Fig. 3 presents the DRIFT spectra of the kaolin and metakaolin in which kaolin displays specific bands at the frequencies of 430/470, 538, 696, 754/789, 912/939, 1009/1032/1115, and 3619/3652/3669/3694 cm^{-1} from the Si-O, Si–O–Al, Al–OH, Si–O–Al, Al–OH, Si-O, and –OH vibrations, respectively (da Costa Gardolinski, 2005; Johnston et al., 2008; Tironi et al., 2012; Rondón et al., 2013; Vegere et al., 2019). In addition, both the Si(Al)–O asymmetrical stretching at ~1000 cm^{-1} and the SiO–H stretching at 3600–3700 cm^{-1} region shifted to a lower wavenumber in metakaolin (Davidovits, 2008). This phenomenon usually indicates successful calcination, as confirmed by NMR results (Fig. 4).

The calcination temperature is usually in the range of 600–800 °C to produce reactive metakaolin. The elevating temperature releases water from the mineral kaolinite ($Al_2O_3{\cdot}2SiO_2{\cdot}2H_2O$), the main species of kaolin clay, and disintegrates the material structure, resulting in an amorphous aluminosilicate ($Al_2O_3{\cdot}2SiO_2$), metakaolinite (Ilić et al., 2010). In a vigorous alkaline solution, metakaolin dissolves rapidly, releasing Al and Si producing e.g., geopolymers, zeolite, illite and other products depending on the ambient conditions (de Rossi et al., 2019). The structures of the kaolin and calcined kaolin (metakaolin) were analyzed by using high-resolution ^{27}Al MS-NMR spectroscopy (Fig. 4). Due to the different coordination of O atoms around an Al atom, the peaks of chemical shifts could provide additional information about kaolin and metakaolin. Since metakaolin is used as a starting material in the production of AAM, it is essential to analyze the coordinated Al species in the structure and to determine which Al species may be activated during alkali activation. Kaolin and metakaolin produce peaks of certain ^{27}Al species in the spectrum, such as Al(IV), Al(V), and Al(VI) (Duxson et al., 2007a; Kim, 2010; Davidovits, 2017). Kaolin produces a single Al(VI) peak at −4.1 ppm, which is assigned to six-coordinated Al in kaolinite. Metakaolin produces an Al(IV) peak at 53.3 ppm, as well as an Al(V) peak at about 29 ppm and an Al(VI) peak at 2.7 ppm, which are assigned to five- and six-fold coordinated Al sites.

3. Results

The aspects of this research are the preparation of porous and mechanically stable AAM via optimized preparation steps/process and the in-depth characterization of Na-AAM and K-AAM. Porosity refers to the presence of many small holes (meso- and micro-pores) that are permeable to water, chemical solutions, and gas (air) for instance, however, increasing porosity typically decrease the strength of the material. As defined by Kumar and Bhattacharjee (2003), porosity is the ratio of the volume of all pores to the volume of a material(Kumar and Bhattacharjee, 2003). Also, the water content plays a role in polymerization reactions not only as a solvent but also as a transport medium for ions; when the H_2O/Al_2O_3 ratio increases, porosity has been noticed to increase(Okada et al., 2009) The effects of the curing parameters on the

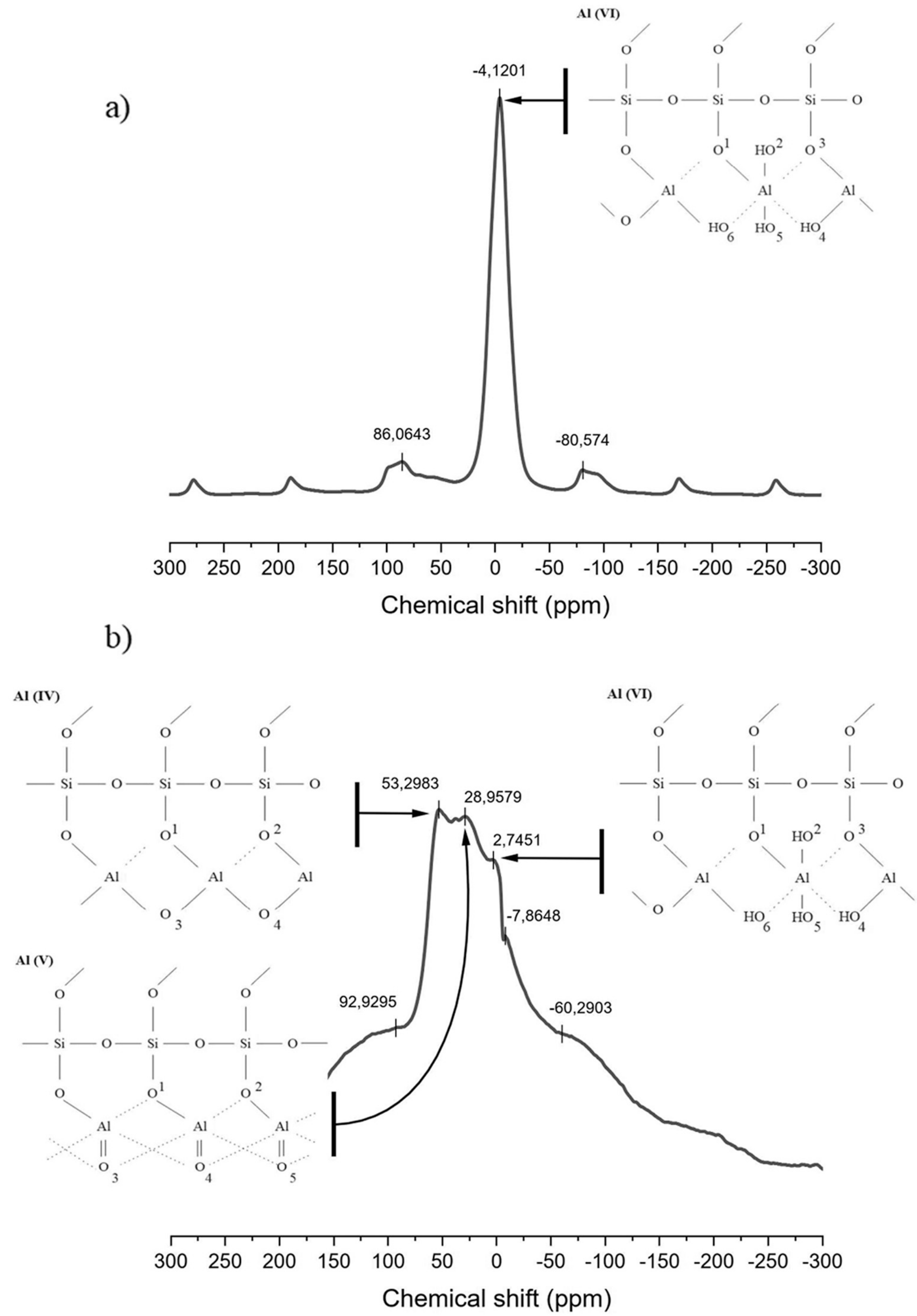

Fig. 4. ^{27}Al MS-NMR spectra of (a) kaolin and (b) calcined kaolin (metakaolin).

AAM volume, viscosity, and BET values, as well as the compressive strength test results are presented in Table 3.

3.1. Viscosity

Viscosity is a property that affects how a material can be further modified, for example, during a manufacturing process. Previous research has highlighted that the viscosity of Na-AAM paste was higher than that of K-AAM paste (Sabitha et al., 2012).

In this study, it was found that the viscosity of the Na-AAM paste was indeed higher than that of the K-AAM paste when the molar ratios for metal oxides and water in Na-AAM and K-AAM were kept the same for comparison. The viscosity of the K-AAM paste was approximately 70% lower than that of the Na-AAM paste (Table 3). To increase the porosity of the AAM, foaming agents were added to the pastes; regardless of the amount or type of the foaming agent (H_2O_2 vs SPC) used, the viscosity variation for both alkalis was nearly the same. With an increasing porogen concentration, the viscosity decreased. The volume increase in the tube correlated with viscosity when H_2O_2 was used as porogen; the higher viscosity of the Na-AAM slurry caused the volume of the Na-AAM paste to increase by half as much as that of the K-AAM paste; this difference may be attributed to the escape of gas bubbles from the K-AAM slurry due to is low viscosity. However, the volume increase was not found to be correlated with BET values, indicating that only the large-

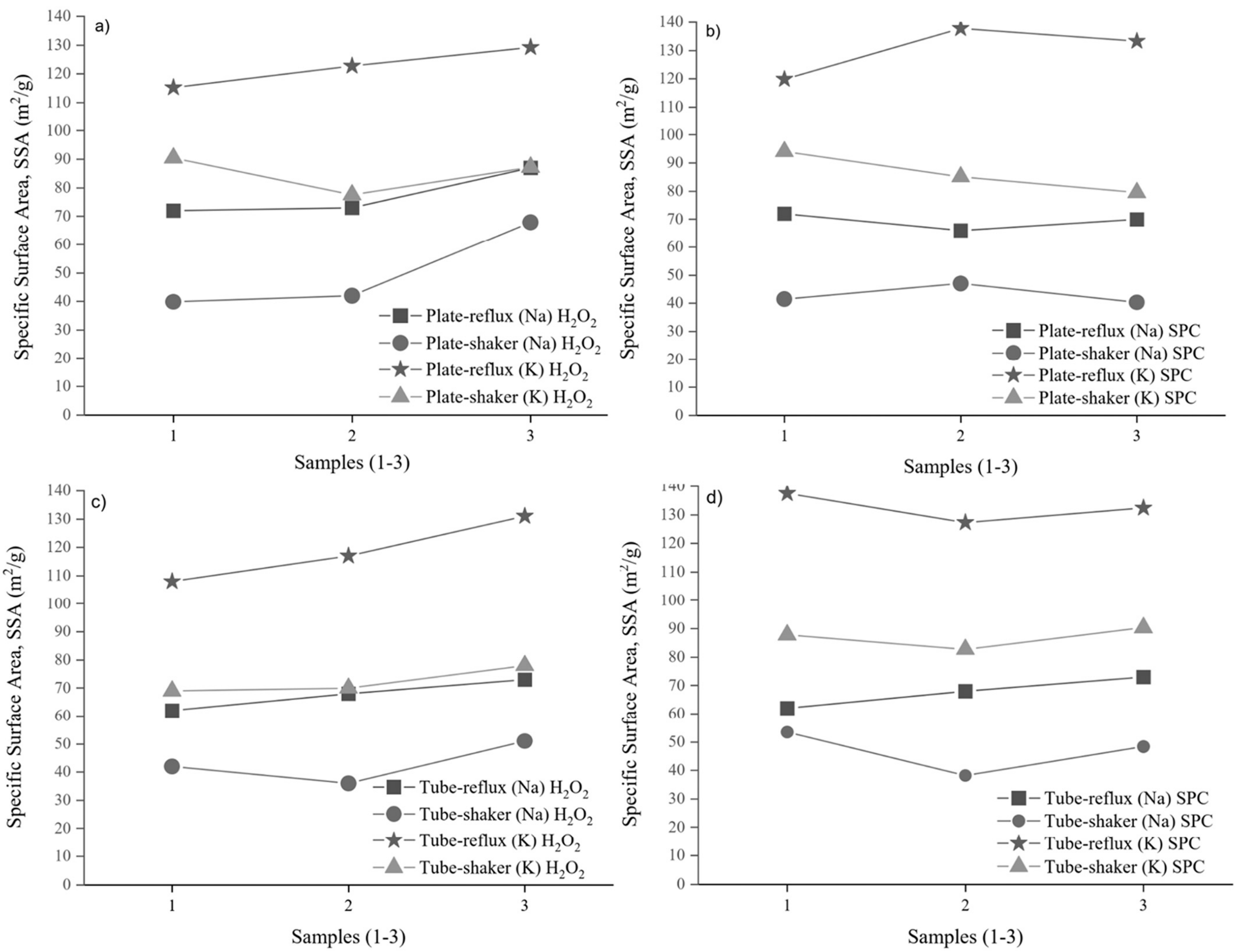

Fig. 5. Effect of the washing methods on the porosity of Na-AAM and K-AAM. The amounts (total mass%) of porogen added in samples 1, 2 and 3 were 0%, 0.1%, and 0.5%, respectively. The materials were denoted as follows: (a) plate H2O2, (b) plate SPC, (c) tube H2O2, and (d) tube SPC.

sized pores (macropores) were greater in size in the Na-AAM slurries than in the K-AAM slurries. This finding was confirmed by FESEM analysis. Furthermore, the analysis of the different alkali activator ratios (Table 4) showed that the optimum ratio of 0.7:14.1 correlated with the highest viscosity values for the Na-AAM slurry. Such a result was not seen in the K-based alkali activator. The alkali ratios (Na_2O/SiO_2:H_2O/Na_2O) of 0.5:14.1 and 0.7:11.0 for Na-AAM were not produced due to the negative water concentration.

3.2. Effect of curing temperature on compressive strength

The porosity and pore size distribution in the microstructure of a material are important properties that affect durability, permeability, and adsorption rate (Hajimohammadi et al., 2011). Compressive strengths were determined from the Na-AAM and K-AAM samples cured both at 60 °C and 105 °C and presented in Tables 3 and 4.

After being subjected to different curing temperatures, the materials appeared considerably similar externally. Moreover, when H_2O_2 was used as porogen, the Na-AAM and K-AAM were equally durable based on compressive strength after being cured at 105 °C.

The Na-AAM samples were compared in terms of the porogen used (H_2O_2 vs SPC) and curing temperature (60 °C vs 105 °C). The SPC imparted greater compressive strength than H_2O_2 as it produced a lower degree of porosity (wherein the volume increase was two to three times lower). As regards curing temperatures, the samples cured at 60 °C displayed compressive strength that is two to four times higher than those cured at 105 °C. At a curing temperature beyond 60 °C, the samples probably contained some water, which rendered them stronger. The compressive strength of metakaolin has been found to be highest at a curing temperature of approximately 80 °C; therefore, 105 °C was probably too high (Kong et al., 2008).

3.3. Effect of the washing procedure on surface area

The Na-AAM and K-AAM samples cured at 105 °C were washed by refluxing at the boiling point of 1 M acetic acid for 24 h. Then, the washed materials were oven-dried at 105 °C for 24 h. For comparison, Na-AAM and K-AAM samples were washed on a laboratory shaker at room temperature for 30 min and oven-dried at 105 °C for 24 h. The SSA was determined by the BET method (Table 3). Fig. 5 shows the effect of the washing method on the SSA of the Na-AAM and K-AAM with both H_2O_2 and SPC as porogens. Based on the results, it can be concluded that refluxing the AAM at the boiling point increased the SSA by approximately 30% compared with washing the materials on a laboratory shaker at room temperature. Because due to the refluxing, the pores of the materials expand and the acid penetrates deeper in the material, it is also very possible that acid is causing erosion not only on the surface of the materials but in the inner parts, effecting on the SSA readings (Bai

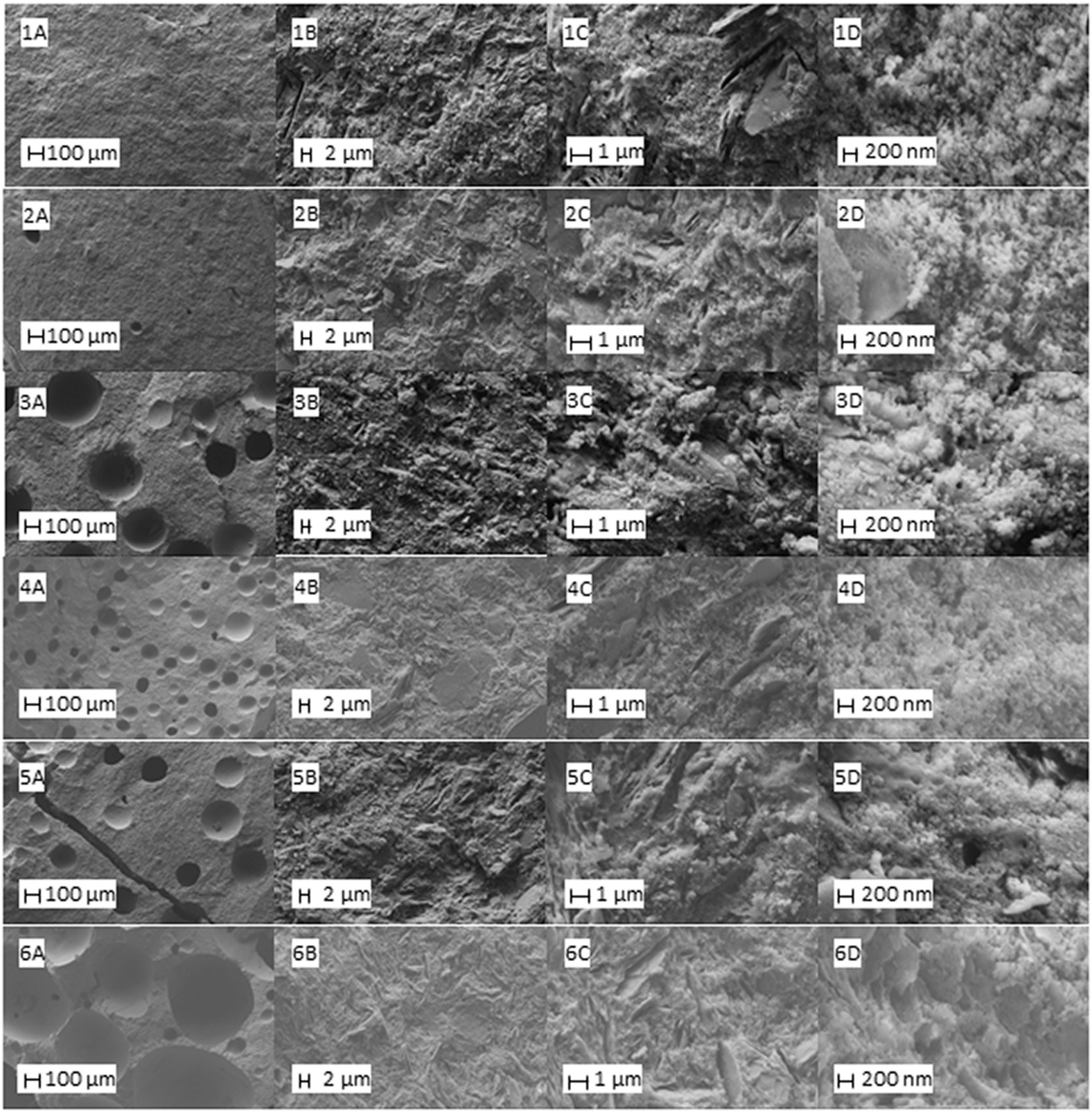

Fig. 6. FESEM images for Na-AAM and K-AAM obtained at different magnifications: (A) 150, (B) 5 K, (C) 20 K, and (D) 75 K. The amount of alkali added to H_2O_2 as porogen (% of the total mass) were (1) K 0%, (2) Na 0%, (3) K 0.1%, (4), Na 0.1%, (5) K 0.5%, and (6) Na 0.5%.

and Colombo, 2018; Bai et al., 2018). Moreover, the results showed that regardless of whether H_2O_2 solution or powdered SPC was used, the SSAs were considerably similar. The SPC did not enhance the production of a porous material.

3.4. FESEM

FESEM images revealed the differences in volume increase between the Na-AAM and K-AAM samples (Table 3, Fig. 6). The significant differences in pore sizes as a function of porogen concentration were observed; in K-AAM, the pore size (~100 µm) did not increase when H_2O_2 concentration was increased from 0.1% to 0.5%, whereas in Na-AAM, an increase from 50 µm to 200–250 µm (magnitude of four to five) was noted. This finding is in line with the sample volume increase during curing (Table 3). However, the increase in volume and pore size did not correlate with BET values; the pore walls were smooth and therefore they do not cause increase in BET values (images indicated with the letter A). As regards morphology, the images obtained at high magnifications (up to 75 K) showed that in the samples, a flake-like structure was found among the micelle-like structures and was identified as unreacted aluminosilicate from metakaolin (images 2C, 2D, 4B, 4C, 6C, and 6D) (Barbosa et al., 2000b; Okada et al., 2009).

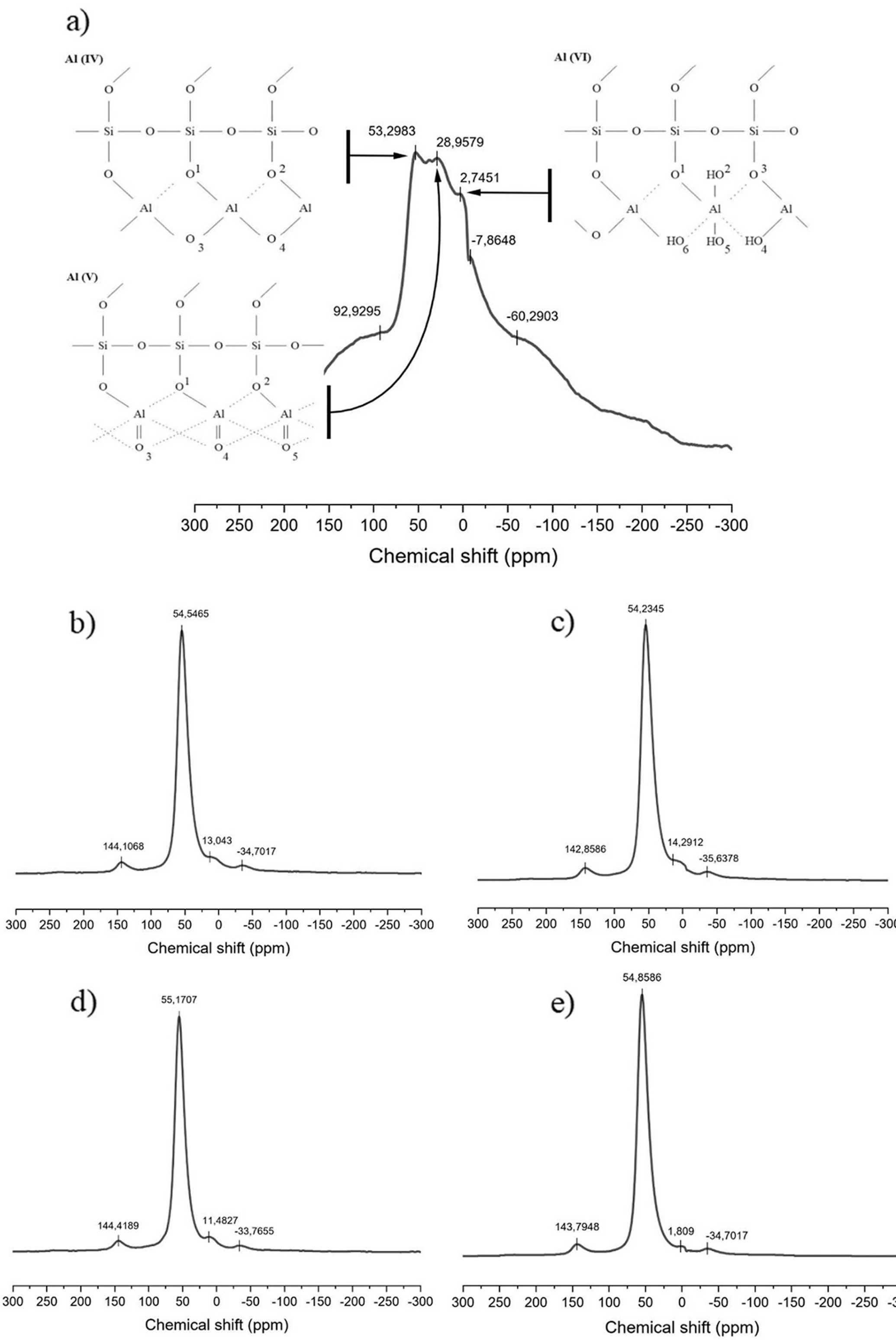

Fig. 7. ^{27}Al MS-NMR chemical shifts in (a) metakaolin, (b) Na-AAM cured at 60 °C, (c) Na-AAM cured at 105 °C, (d) K-AAM cured at 60 °C, and (e) K-AAM cured at 105 °C.

3.5. ^{27}Al MS-NMR

High-resolution ^{27}Al MS-NMR is a useful technique in studying the structures of silicates and aluminosilicates due to the different coordination of O atoms around an Al atom. ^{27}Al MS-NMR spectra were measured from kaolin, metakaolin, and Na-AAM and K-AAM cured at 60 °C and 105 °C (Fig. 7).

Metakaolin produces peaks of certain ^{27}Al species in the spectrum, such as Al(IV), Al(V), and Al(VI), with a chemical shift of 53.3, 29, and 2.7 ppm, respectively (Lizcano et al., 2012b; Davidovits, 2017). Metakaolin, Na-AAM, and K-AAM produce identical Al(IV) peaks at approximately 54 ppm due to their tetrahedral Al sites. Metakaolin also

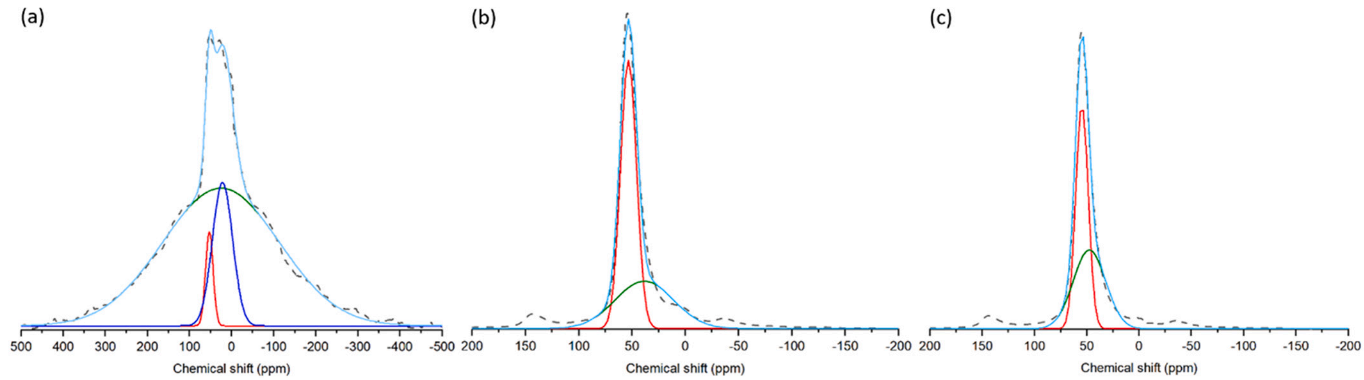

Fig. 8. ^{27}Al NMR spectrum of (a) metakaolin, (b) Na-AAM and (c) K-AAM deconvolution into Al(IV), Al(V), and Al(VI) peaks.

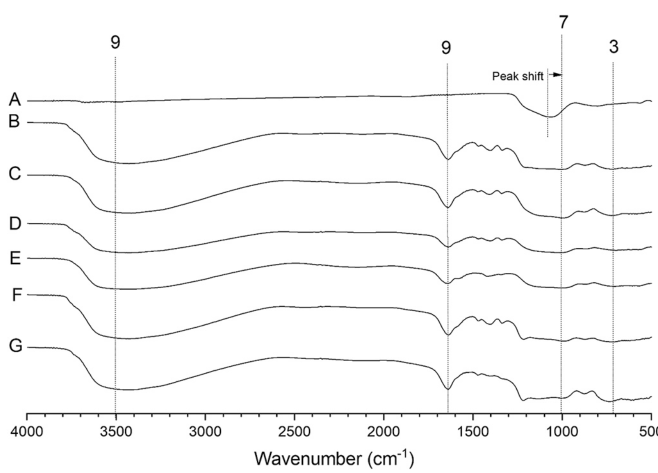

Fig. 9. FTIR spectra of selected AAM: (a) Metakaolin, (b) K-AAM 0%, (c) Na-AAM 0%, (d) K-AAM 0.1%, (e) Na-AAM 0.1%, (f) K-AAM 0.5%, and (g) Na-AAM 0.5% (% indicates the amount of H_2O_2 as porogen, mass%).

produces an Al(V) peak at approximately 29 ppm and an Al(VI) peak at 2.7 ppm related to five- and six-fold coordinated Al sites. Xu Chen et al. (2017) have reported that during the alkalinization the Al(IV) was increased and both Al(V) and Al(VI) were decreased as a function of time. This is usually observed during the polymerization and indicating dissolution of metakaolin (Chen et al., 2017). As can be seen from the spectrum, the Al (V) and Al (VI) signals of the Na-AAM and K-AAM are absent at 29 and 2.7 ppm. The same can be seen in the spectra (Fig. 8), where is presented metakaolin, Na-AAM and K-AAM deconvolution into Al(IV), Al(V), and Al(VI) peaks. Al(IV) has almost been unchanged during the polymerization, but the proportion of Al(V) has increased whereas Al(VI) has vanished. Based on this finding, it can be stated that Al(V) and Al(VI) species have participated in the polymerization reaction and that the Al-O coordinates have changed. Meanwhile, the strong signal given by Al(IV) indicates that the Al species in question do not participate in the polymerization reaction (Davidovits, 2017). Al(V) has a tense configuration due to Al═O double bond, which in turn reacts easily in alkali silicate solutions. Al(IV) and Al(VI) activity would require violation of individual Al-O individual bonds, which are stable. Thus, these reactions occur much slower than Al (Samuel et al., 2021). As described above, this study found that, in addition to Al(V), Al(VI) was also quite reactive because this signal was completely missing from the Na and K-AAM ^{27}Al NMR spectra. Table 5 presents ^{27}Al NMR chemical shifts (ppm) and relative proportions (%) for coordinated Al. OriginPro software (2020b (64-bit) 9.7.5.184 (Academic) Copyright © 1991–2020 OriginLab Corporation) was used to fit spectra and determine the relative proportions for coordinated Al.

Table 2
Vibrational modes of FTIR for kaolin, metakaolin, and AAM.

Vibrational mode	Wavenumber (cm^{-1})	Corresponding number in Figs. 3 and 7	References
Si–O–Si rocking	450–490	1	(Tironi et al., 2012; Z. Zhang et al., 2012a; Vegere et al., 2019)
Si(Al)–O–Al bending	540	2	(Barbosa et al., 2000b; Tironi et al., 2012; Vegere et al., 2019)
Si–O symmetrical stretching/ Al–OH	694	3	(Barbosa et al., 2000b; Tironi et al., 2012; Vegere et al., 2019)
Al–O (4-coordinated Al bending)	810	4	(Rahier et al., 1997; Ríos et al., 2009a)
Si–O–Al (kaolin)	754, 789		(Tironi et al., 2012)
Si–O–Si bending Si–OH bending	810–840	5	(Barbosa et al., 2000b)
Al–OH–Al (6-coordinated Al, OH bending)	914–918	6	(Barbosa et al., 2000b; Ríos et al., 2009b; Tironi et al., 2012)
Al–OH (kaolin)	939		(Tironi et al., 2012)
Si–OH symmetrical stretching	900–980	7	(Aguiar et al., 2009)
Al–O stretching (tetrahedral AlO_4)	990–1020	7	(Ríos et al., 2009a)
Si(Al)–O asymmetrical	1008	7	(Barbosa et al., 2000a; Vegere et al., 2019)
Si–O stretching symmetrical	993–1100	8	(Rahier et al., 1997; Barbosa et al., 2000b; Tironi et al., 2012)
H_2O molecular Al/Si–OH···H_2O	1620–1640 3350–3500	9	(Rahier et al., 1997; Barbosa et al., 2000b)
SiO–H stretching vicinal or geminal	~3660	10	(Aguiar et al., 2009)
SiO–H stretching isolated	~3750	11	(Aguiar et al., 2009)

3.6. Drift

The band shift in DRIFT spectra, which indicates amorphous nature of the AAM and success of geopolymerization (asymmetric Si–O–Si/Al–O–Si bands of ~1080–1000 cm^{-1}), can be seen from the spectrum

Table 3
Effects of foaming agents and curing parameters on the AAM volume, viscosity, and BET values, along with the compressive strength test results (max = over the maximum measurement capacity of equipment (10 MPa), possible measurement error marked with asterisk*).

Sample	Alkali ratio X_2O/ SiO_2:H_2O/ X_2O	H_2O_2 as foaming agent (mass-%)	Volume increase in tube (%)	Viscosity (Pa)	BET plate reflux (m^2/g)	BET plate shaker (m^2/g)	BET tube reflux (m^2/g)	BET tube shaker (m^2/g)	Strength 60 plate (MPa)	Strength 105 plate (MPa)	Strength 60 tube (MPa)	Strength 105 tube (MPa)
K-AAM	0.7:14.1	0.0	0.0	2.064	107	84	108	69	–	6.1	–	4.9
H_2O_2	0.7:14.1	0.1	17.0	1.720	114	72	117	70	–	6.0	–	2.9
	0.7:14.1	0.2	20.0	1.744	–	–	–	–	–	3.4	–	1.2
	0.7:14.1	0.3	33.3	1.650	–	–	–	–	–	2.2	–	1.0
	0.7:14.1	0.4	33.3	1.712	–	–	–	–	–	1.1	–	1.0
	0.7:14.1	0.5	20.0	1.720	120	81	131	78	–	0.7	–	0.9
Na-	0.7:14.1	0.0	0.0	6.943	72	37	62	42	7.9	7.6	9.0	5.2
AAM	0.7:14.1	0.1	13.6	5.519	73	39	68	36	7.6	3.4	9.0	1.1*
H_2O_2	0.7:14.1	0.2	35.0	5.487	–	–	–	–	6.9	4.1	2.3*	2.4
	0.7:14.1	0.3	37.2	5.479	–	–	–	–	1.9	1.6	6.3	1.0*
	0.7:14.1	0.4	60.0	5.325	–	–	–	–	1.1	1.0	5.1	1.8
	0.7:14.1	0.5	69.0	4.999	87	63	73	51	0.4	0.4	3.6	1.7
K-AAM	0.7:14.1	0.0	0.0	1.942	107	84	108	69	–	–	–	–
SPC	0.7:14.1	0.1	11.1	1.814	123	76	100	65	–	–	–	–
	0.7:14.1	0.2	14.3	1.824	–	–	–	–	–	–	–	–
	0.7:14.1	0.3	26.2	1.800	–	–	–	–	–	–	–	–
	0.7:14.1	0.4	30.2	1.808	–	–	–	–	–	–	–	–
	0.7:14.1	0.5	40.5	1.740	119	71	104	71	–	–	–	–
Na-	0.7:14.1	0.0	0.0	6.943	72	37	62	42	max	7.6	9.0	5.2
AAM	0.7:14.1	0.1	4.5	5.479	66	42	68	30	max	max	8.1	2.0
SPC	0.7:14.1	0.2	8.9	5.263	–	–	–	–	max	max	3.6*	1.0
	0.7:14.1	0.3	25.6	5.600	–	–	–	–	max	4.2	3.6*	1.0
	0.7:14.1	0.4	31.8	5.287	–	–	–	–	max	4.7	5.7	1.2
	0.7:14.1	0.5	37.5	5.487	70	36	72	38	max	3.4	4.8	0.3

Table 4
Curing, viscosity and compressive strength test results for Na-AAM and K-AAM prepared using different alkali ratios.

Sample	Alkali ratio X_2O/ SiO_2: H_2O/X_2O	H_2O_2 as foaming agent (mass-%)	Tube volume increase (%)	Viscosity (Pa)	Strength 60 plate (MPa)	Strength 105 plate (MPa)	Strength 60 tube (MPa)	Strength 105 tube (MPa)
	0.5:14.1	0.3	37.5	2.999	–	–	–	–
	0.6:14.1	0.3	27.7	1.632	–	–	–	–
	0.7:14.1	0.3	33.3	1.650	–	–	–	–
	0.8:14.1	0.3	22.5	1.336	–	–	–	–
K-AAM H_2O_2	0.9:14.1	0.3	25.0	1.570	–	–	–	–
Modifications	0.7:11.0	0.3	22.0	2.391	–	–	–	–
	0.7:12.5	0.3	31.0	1.824	–	–	–	–
	0.7:14.1	0.3	33.3	1.650	–	–	–	–
	0.7:15.5	0.3	33.3	1.656	–	–	–	–
	0.7:17.0	0.3	26.1	1.460	–	–	–	–
	0.5:14.1	0.3	–	–	–	–	–	–
	0.6:14.1	0.3	40.0	2.152	4.3	1.4	5.4	1.7
	0.7:14.1	0.3	37.2	5.479	1.9	1.6	6.3	1.0
	0.8:14.1	0.3	20.0	0.752	2.5	3.4	5.9	3.7
Na-AAM H_2O_2	0.9:14.1	0.3	17.5	0.480	3.0	4.1	3.8	2.4
Modifications	0.7:11.0	0.3	–	–	–	–	–	–
	0.7:12.5	0.3	42.5	2.463	2.2	2.8	3.7	3.3
	0.7:14.1	0.3	37.2	5.479	1.9	1.6	6.3	1.0
	0.7:15.5	0.3	23.8	0.728	3.4	4.9	8.7	3.5
	0.7:17.0	0.3	11.6	0.472	2.1	4.0	3.3	2.4

Table 5
^{27}Al NMR chemical shifts (ppm) and relative proportions (%) for coordinated Al in (a) metakaolin, (b) Na-AAM and (c) K-AAM.

	Metakaolin			Na-AAM			K-AAM		
Coordinated Al	(IV)	(V)	(VI)	(IV)	(V)	(VI)*	(IV)	(V)	(VI)*
Chemical shift (ppm)	53.3	29.0	2.75	54.4	13.7	–	55.1	6.66	–
Relative proportion (%)	62	27	11	53	47	–	60	39	–

* Al(VI) chemical shift (ppm) was not detected.

(Fig. 9) (Mackenzie et al., 2007; Davidovits, 2008; Vegere et al., 2019).

The small band at around 720 cm^{-1} indicates that tetrahedral Al is the main Al species, as confirmed by NMR data (Z. Zhang et al., 2012b). The band at around 1450 cm^{-1} is suggested to correspond to the O–C–O stretching in carbonates (Juhola et al., 2019; Vegere et al., 2019). However, the molecular water in Al/Si–OH⋯H_2O can clearly be seen with bands at around 1630 and 3500 cm^{-1} (Table 2).

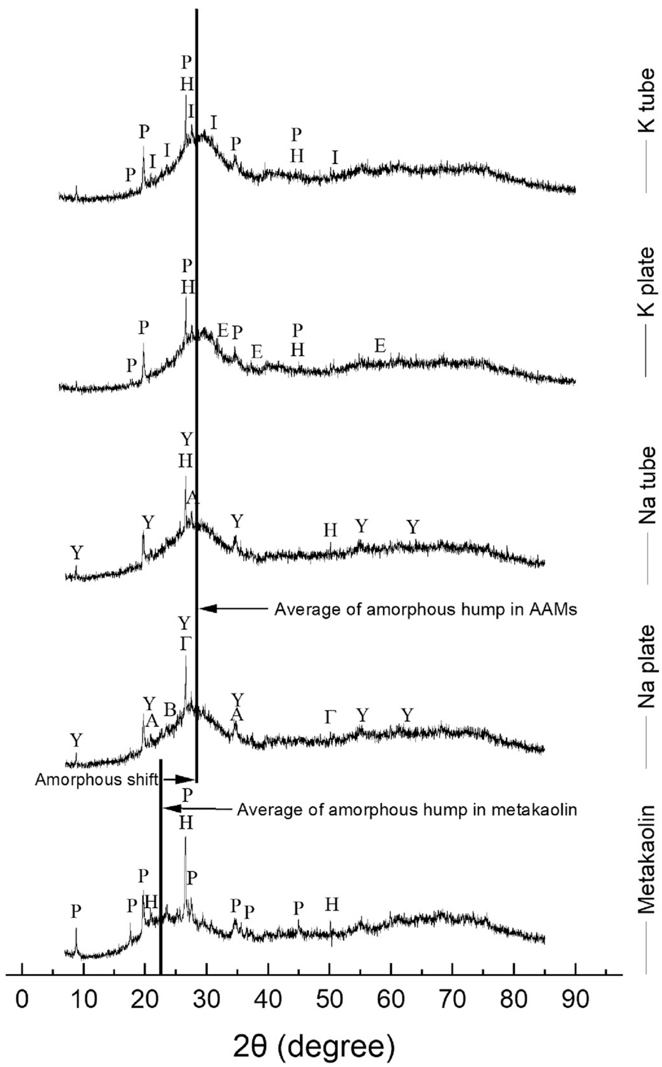

Fig. 10. X-ray diffractograms of metakaolin vs Na-AAM and K-AAM. Legend: H = hexagonal silicon oxide (quartz), P = potassium aluminum silicate, B = aluminum silicate, A = sodium aluminum silicate, Y = aluminum silicate hydroxide hydrate (halloysite), E = tetragonal potassium aluminum oxide, and I = potassium aluminum silicate (microcline).

3.7. XRD

The comparison between the diffractograms of metakaolin and AAM revealed the shift of the amorphous hump from 2θ values of 23 to 28 (Fig. 10). This phenomenon is identified as structural changes in aluminosilicates during polymerization (Duxson et al., 2007b). As has been reported in the literature, Na-AAM contains water in hydrate form (halloysite) compared with K-AAM (Waijarean et al., 2014). This is due to different behavior of Na^+ and K^+ ions; Na^+ ions are smaller (ionic radius: 0.102 vs 0.138), and they are more electronegative than K^+ ions (0.93 vs 0.82 based on the Pauling scale) thus, they attract more OH^- ions resulting in the increased water content (chemically bonded) of Na-AAM (Lizcano et al., 2012a). Furthermore, the formation of crystalline aluminosilicate phases in all AAM was noticeable.

For the Na-AAM plate (Fig. 10), the X-ray diffractogram reflections were matched with sodium aluminum silicate ($Na_{0.775}Al_{0.775}Si_{0.225}O_2$; ICDD 04–010-3960), and aluminum silicate ($AlSi_{0.5}O_{2.5}$; ICDD 00–029-0084). For the Na-AAM tube, the reflection were matched with sodium aluminum silicate ($NaAl_3Si_3O_{11}$; ICDD 00–046-0740). Aluminum silicate hydroxide hydrate ($Al_2Si_2O_5(OH)_4{\cdot}2H_2O$; ICDD 00–029-1489) was the common phase for the Na-AAM plate and the Na-AAM tube. For the K-AAM plate and K-AAM tube, the reflections were matched with potassium aluminum oxide ($KAlO_2$; ICDD 04–016-3974) and potassium aluminum silicate ($KAlSi_3O_8$; ICDD 01–071-0955), respectively. The common phase for the K-AAM plate and the K-AAM tube was potassium aluminum silicate ($KAl_3Si_3O_{11}$; ICDD 00–046-0741). Hexagonal silicon oxide (SiO_2; ICDD 00–046-1045) reflection was present in every sample.

4. Conclusion

In this study, two alkaline activators (Na and K) with different X_2O/SiO_2, SiO_2/Al_2O_3, X_2O/Al_2O_3, and H_2O/X_2O (X = Na, K) ratios in metakaolin-based AAM were investigated. The results demonstrated the following:

(a) The viscosity of the K-AAM paste was 70% lower than that of the Na-AAM paste. However, the volume of the Na-AAM paste in airtight tube was three times higher than that of K-AAM, but the BET values of K-AAM were 30% higher than those of Na-AAM.
(b) Using H_2O_2 as porogen, the produced Na-AAM and K-AAM were equally durable after being cured at 105 °C. Using SPC imparted greater compressive strength than H_2O_2, as it created a lower degree of porosity (wherein the volume increase was two to three times lower). Moreover, compressive strength was higher by two to four times in the materials cured at 60 °C than in those cured at 105 °C.
(c) SSA remained the same regardless of the foaming agent used and regardless of the curing vessel method used (open muffin pan or in sealed plastic tube).
(d) The FESEM images of the Na-AAM and K-AAM produced with H_2O_2 showed significant differences in pore sizes as a function of porogen concentration; in K-AAM, the pore size (~100 μm) did not increase when H_2O_2 concentration was increased from 0.1% to 0.5%, whereas in Na-AAM, an increase from 50 μm to 200–250 μm (magnitude of four to five) was noted. The higher SSA values of the K-AAM were due to its smaller pore size; in Na-AAM, the increase in porogen concentration only increased the pore size.
(e) ^{27}Al MS-NMR, FTIR, XRF, and XRD analyses revealed the successful preparation of metakaolin through the calcination of kaolin at 750 °C. Furthermore, based on the ^{27}Al MS-NMR and FTIR results, the AAM prepared from the produced metakaolin demonstrated the transformation of Al species, that is, from Al (*VI*), Al(V), and Al(IV) to Al(IV), confirming the full polymerization of metakaolin into AAM. The polymerization of metakaolin was also noticed based on the amorphous band and reflection shifts in both the DRIFTS and XRD data.
(f) The XRD data showed that the Na-AAM contained water in hydrate form (halloysite) compared with the K-AAM, suggesting the different polymerization reaction rates between these AAM.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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